

# Nandhini Kasukurthi

## Data Engineer

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## Education

Master of Science in Computer Science  
University of North Texas, Denton, Texas

August 2023 – May 2025

Bachelor of Technology in Computer Science  
Gitam University

June 2018 – May 2022

## Technical Skills

- **Programming Languages:** Python, Scala, R, C++, C, Java, Shell Scripting
- **Web Technologies:** HTML, CSS, React JS
- **Databases:** Oracle, PL/SQL, Azure SQL Warehouse, Azure SQL DB, Azure Cosmos NO SQL DB, Teradata, Vertica, RDBMS, MySQL, Oracle, PostgreSQL, Microsoft SQL Server
- **Big Data Technologies:** Spark, PySpark, Databricks, Informatica PowerCenter
- **Hadoop Eco System:** Spark, Hive, Sqoop, Oozie, Pig
- **Data Visualization Tools:** Power BI, Tableau
- **AWS Cloud Platform:** AWS EC2, AWS RDS, AWS S3, AWS Lambda, AWS SNS, AWS VPC, AWS Kinesis, AWS IAM, AWS Dynamo DB, AWS SQS, AWS Athena, AWS QuickSight
- **Platforms:** Linux, Windows and OS X

## Professional Experience

Tata Consultancy Services  
Assistant System Engineer

July 2022 – July 2023

- Directed the design and implementation of a microservices framework using FastAPI, enhancing application functionality and system robustness.
- Developed and integrated a real-time environmental monitoring solution leveraging Swagger API, enabling prompt generation and dispatch of Departure Notification reports with 20% improvement in accuracy and timeliness.
- Composed detailed, user-friendly technical documentation that streamlined communication and accelerated onboarding for internal teams and clients.
- Managed the complete microservices development lifecycle, applying RESTful API principles and containerization techniques, resulting in 15% faster project delivery and improved deployment efficiency.

Caremotto  
Data Analyst

July 2021–June 2022

- Led a pivotal microservices project using FastAPI, significantly improving application functionality with innovative development approaches, resulting in a 67% performance boost  
Monitored real-time pollution factors, showcasing environmental responsibility.
- Implemented an efficient system with Swagger API and microservices for prompt generation and dispatch of Departure Notification reports.
- Took a pivotal role in meticulously crafting extensive and user-friendly documentation.
- Ensured adherence to high standards, facilitating better understanding for both internal teams and clients.
- The commitment to clear and comprehensive documentation contributed to seamless communication and project understanding.

## Projects

Coffee Store Sales Prediction

- Conducted comprehensive analysis of sales data for an international coffee chain, employing linear regression and advanced ensemble models such as Random Forest, Gradient Boosting, and XGBoost.
- Processed Coffee Sales dataset using Excel for data manipulation, Power BI for visualization, and SQL for querying, improving data processing efficiency by 15%.
- Applied hyperparameter optimization and predictive analytics, delivering actionable insights that enhanced inventory management and reduced waste.
- Executed in-depth exploration of sales trends, utilizing advanced statistical techniques to forecast sales with accuracy, supporting strategic planning and revenue optimization.

## Real-Time Data Pipeline for E-Commerce Analytics

- Designed and implemented real-time event ingestion system capturing user interactions such as page views, cart updates, and purchases, ensuring data integrity and reliability.
- Developed streaming aggregation jobs processing daily events, generating key metrics such as daily sales and top-selling products with 15% latency reduction.
- Architected separate storage layers for raw and processed datasets using distributed file systems and NoSQL databases, improving query performance.
- Created interactive dashboard visualizing KPIs, enabling real-time monitoring of sales trends and user behavior, driving data-driven decisions with 20% faster insights.

## Real-Time Streaming Data Pipeline Using Kafka and Spark

*Technologies: Apache Kafka, Apache Spark Streaming, AWS EMR, PySpark*

- Designed and implemented a scalable streaming data pipeline for processing continuous JSON events from multiple source systems via Kafka.
- Utilized Spark Structured Streaming on AWS EMR clusters to perform real-time data transformations and store results in Parquet format on AWS S3 for downstream analytics.
- Optimized cluster resource usage by implementing autoscaling and efficient batch interval settings, reducing processing latency by 30%.

## Azure Data Factory ETL Pipeline for Multi-Source Data Integration

*Technologies: Azure Data Factory, Azure Databricks, PySpark, Azure Data Lake Storage*

- Architected and developed data ingestion pipelines using ADF v2 to extract, transform, and load data from disparate relational and NoSQL sources into Azure Data Lake Storage.
- Created reusable PySpark notebooks and data flows in Databricks to perform complex data transformations and maintain high data quality.
- Automated workflows using ADF triggers (schedule, event, tumbling window) and integrated Azure Logic Apps for orchestration of business processes.

## Big Data Analytics Platform with AWS Glue and Redshift

*Technologies: AWS Glue, AWS Redshift, Python, Spark*

- Developed ETL workflows using AWS Glue Spark jobs to clean, aggregate, and transform large datasets stored in AWS S3 for analytics in Redshift.
- Created Python and Spark scripts for data preprocessing and feature engineering to support machine learning use cases.
- Integrated Glue jobs with Jenkins and AWS Lambda to enable automated, serverless orchestration and pipeline scheduling.

## Certifications

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- Attained AZ-900 Microsoft Certification.
- Completed Database Management Essentials – Coursera.
- Attained Google Cloud Professional Data Engineer Certification.
- Achieved AWS Certified Data Analytics.
- Certified in Databricks Data Engineer Associate.